

Marwan Abudaif

Cairo, Egypt • +20 155 292 9790 • marwanabudaif5@gmail.com

[LinkedIn](#) | github.com/MarwanMoh11 | Open to: New Grad SWE • Egypt / UAE / Remote EMEA • Dec 2026

SUMMARY

Computer Engineering senior (GPA 3.48/4.0, AUC) shipping production software at METI Software, where I am the sole engineer on a multi-tenant Laravel + Next.js client portal and own the chat, dataset and settings surfaces of a geospatial AI platform. Outside work: a 45k-line real-time 3D browser game with WebRTC multiplayer, a native iOS app, and a senior thesis measuring LLM sycophancy.

EDUCATION

B.S. in Computer Engineering

The American University in Cairo (AUC)

Expected Dec 2026

Cairo, Egypt

- GPA 3.48/4.0 • Coursework: Algorithms, Operating Systems, Computer Architecture, Databases, Machine Learning, Distributed Systems • Study abroad: Universidad Carlos III de Madrid (Spring 2025).

SKILLS

- **Languages:** TypeScript / JavaScript, Python, PHP, C++, Swift, SQL, Kotlin, HTML5 / CSS3
- **Frontend:** React 19, Next.js 16, Angular 22, Svelte 5, Tailwind v4, Three.js, shadcn / Base UI
- **Backend & Data:** Laravel 12, Node.js, FastAPI, PostgreSQL + PostGIS, MySQL, Firebase, REST, SSE, WebSockets, WebRTC
- **Infra, Quality & AI:** Docker, AWS, Bitbucket Pipelines CI, Playwright, Vitest, PHPUnit, Git • LLM evaluation & harness design, RAG, MCP agent tooling, Claude & Gemini APIs
- **Certification:** AZ-900 Azure Fundamentals • **Work authorization:** Egypt + UAE residency

EXPERIENCE

Software Engineer

METI Software (Micro Engineering Technologies)

Jun 2025 – Present

Remote, Canada

- Built core surfaces of the **AI Author Tool** (Angular), which parses uploaded documents into structured courses: the Lesson → Sections → Bricks hierarchy, drag-and-drop reordering, content bricks, English/Arabic localisation with RTL, and an SSE stream for course-generation progress.
- On **Aqtar**, a Saudi-facing geospatial services marketplace, shipped the Ratings & Reviews module and the content reporting and moderation system end to end (195 commits, ten-engineer team), plus the English/Arabic localisation layer.
- Promoted from Frontend Intern to **sole engineer** on the **MVS Client Portal**, a multi-tenant Laravel 12 + Next.js 16 product with nine features and three roles: 30k lines of PHP, 35k of TypeScript, 30 migrations, 49 test classes.
- Designed an engagement-health score that is computed on read instead of stored, and cut a naive 21-query-per-screen version down to 3 queries across a 6-month trend.
- Enforced role-scoped authorization, returning 404 instead of 403 so record IDs can't be probed, and tested every route against every role in Playwright to confirm each rule actually applies.
- On **GuloGulo**, a geospatial AI platform (Next.js 16 / React 19 / Laravel / PostGIS), owned chat, datasets and settings (45 commits, 8 merged pull requests). Built a resumable multipart upload engine for 40 GiB archives (batched part signing, bounded concurrency, per-part retry) over an SSE transport that falls back to polling with full-jitter backoff.

Software Engineering Intern

Pulse (formerly Giza Systems), BOOKSHELF platform

Aug 2026

Cairo, Egypt

- Sole author of the delivered Angular 22 application (~3.8k lines): customer catalogue and admin workspace with signal-based state, typed reactive forms, and role guards over a six-route access matrix.
- Wrote the spec the team built against (versioned REST contract, consistent error envelope, security decisions) and met the accessibility and 390/1024/1440px responsive requirements.

Fixed Networks R&D Intern

Nokia & WE Telecom, exploratory placement on FTTH predictive diagnostics (Python, simulator data)

Aug 2025

Cairo, Egypt

FEATURED PROJECTS

GEM Tours (Project Horus): AI Guide for the Grand Egyptian Museum

2026

Next.js 16 • Vertex AI • Firestore vector search • Gemini • Capacitor

- Built solo (18k lines, 72 TypeScript modules): point a phone at an artifact and hear its history in your language. Vertex AI multimodal embeddings in a Firestore vector index identify it in **under 300 ms**. Detection runs in a web worker so the camera preview never stalls, and EN/AR translation with RTL happens at scan time.
- Designed around inference cost: a three-tier cache ending in a shared Firestore store lets one visitor's generated content serve every later visitor, with an on-device TTS fallback. Documented cost is about \$0.008 per visitor.

GeoFighters 2.0: Browser-Native Co-op 3D Shooter

[Play](#) • [GitHub](#)

TypeScript • Three.js • Svelte 5 • Rapier • miniplex ECS • WebRTC

- 45k lines: miniplex ECS with instanced rendering, Rapier physics, PWA and Capacitor iOS builds. Steadied frame time with instancing, spatial-hash separation and a pre-work frame limiter (160 → 59 fps).
- Designed host-star **WebRTC** networking: 30 Hz snapshots on unreliable, unordered SCTP channels so a dropped packet never holds up the next one; per-peer fallback to a socket relay when ICE fails; and TURN on 443/TCP after finding that CGNAT is the norm on Egyptian and UAE carriers.

TELOS II: Senior Thesis Measuring LLM Sycophancy

Aug 2026 – Present

FastAPI • Python • React • TypeScript • Siemens-sponsored • Leading a team of 7

- Designed the system so a versioned, SHA-256-pinned policy behind an independent guard chooses every action, and the LLM only writes the response.